Inside Living Things

A Science A–Z Life Series
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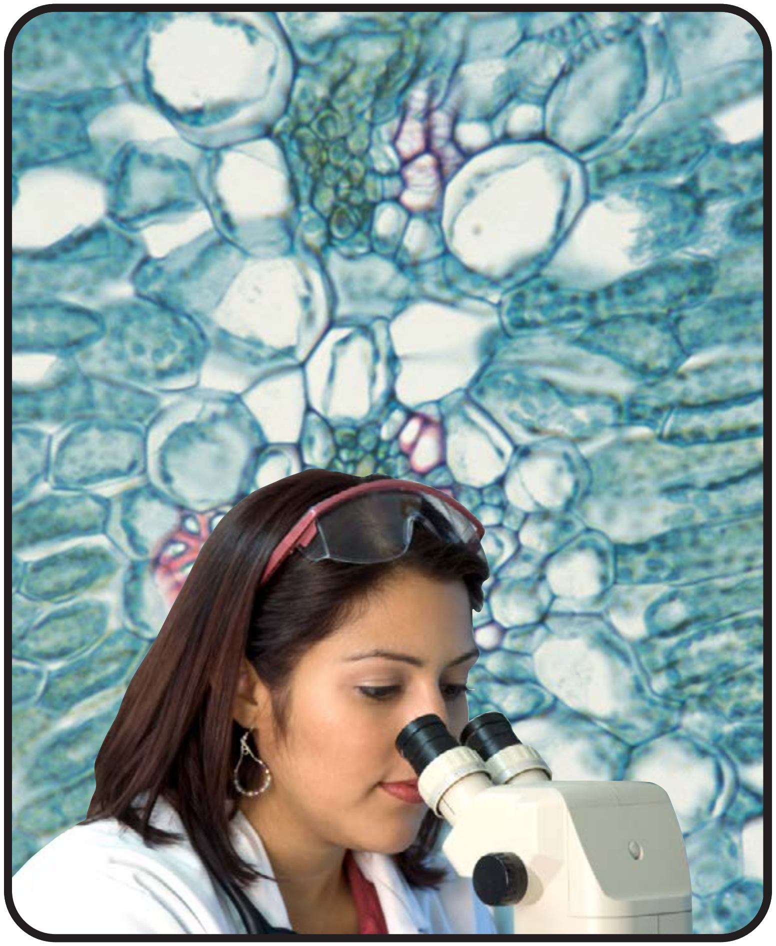

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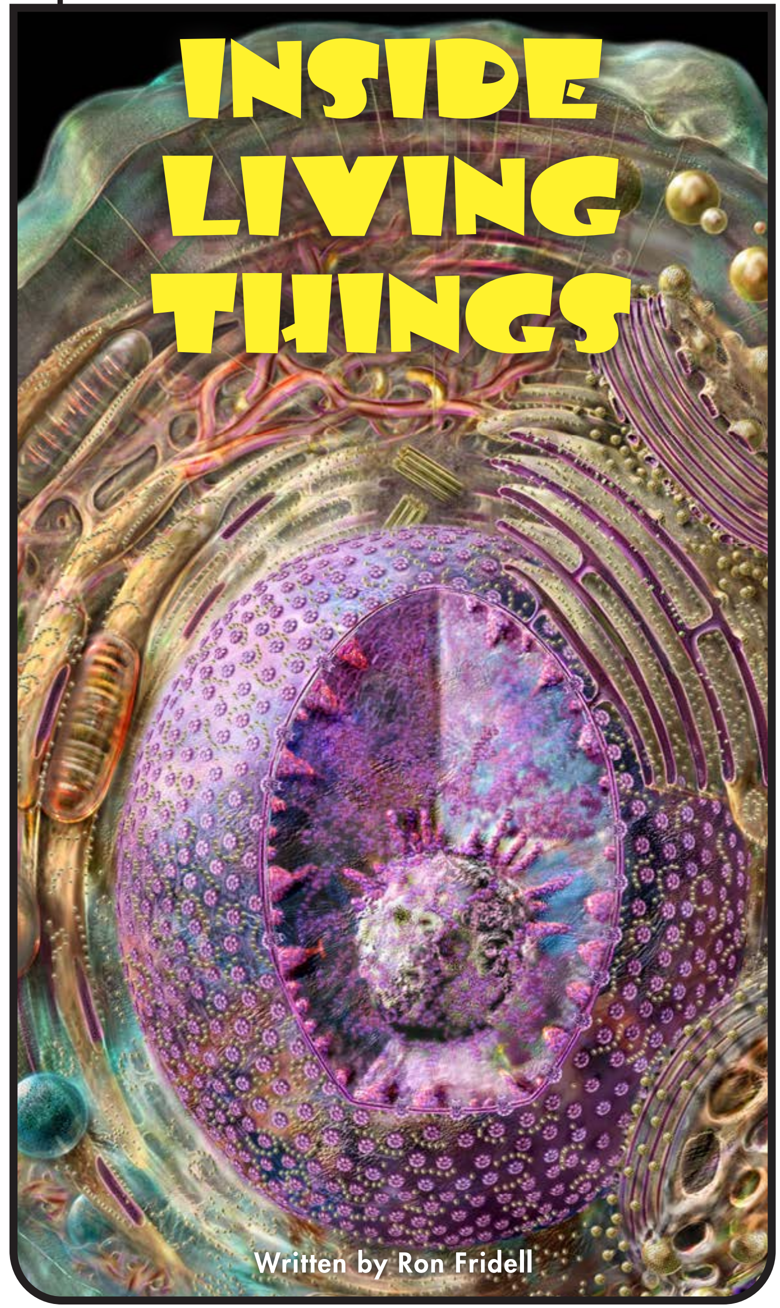

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Written by Ron Fridell

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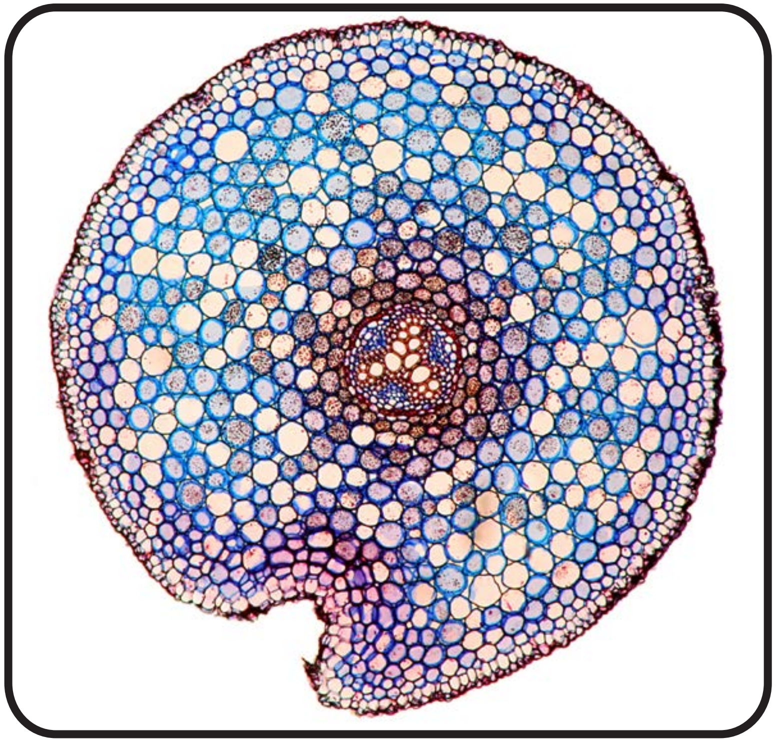

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KEY ELEMENTS USED IN THIS BOOK
The Big Idea: Humans have a common bond with all other life on Earth. All living things are made up of cells and have parts that help them meet their needs. Cells come together to form tissues, tissues work together to make organs, and organs combine to create body systems. These parts must work together to keep an organism healthy. All organisms are susceptible to illness and injury. An understanding of how bodies work can raise our awareness of our own health, leading us toward safe and healthy practices. In this way, we can protect our most important asset—our body.
Key words: antibodies, arteries, bacteria, body system, capillaries, cell, cell membrane, cell wall, chlorophyll, chloroplasts, circulatory system, cytoplasm, digestive system, disease, heart, immune system, infection, lungs, microorganism, mitochondria, multicellular, muscle, muscular system, nervous system, nucleus, nutrients, organ, oxygen, pathogens, photosynthesis, plasma, pores, respiratory system, skeletal system, tissue, trillion, unicellular, vacuoles, veins, viruses **Key comprehension skill:** Compare and contrast *Other suitable comprehension skills:* Main idea and details; cause and effect; classify information; identify facts; elements of a genre; interpret charts, graphs, and diagrams **Key reading strategy:** Connect to prior knowledge *Other suitable reading strategies:* Summarize; ask and answer questions; visualize; using a table of contents and headings; using a glossary and bold terms

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Front cover photo: an illustration of an animal cell
Back cover photo: a scientist looking at a slide of plant cells
Title page photo: a microscopic image of the cross section of the root of a buttercup plant

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Microscopes help us see inside cells.

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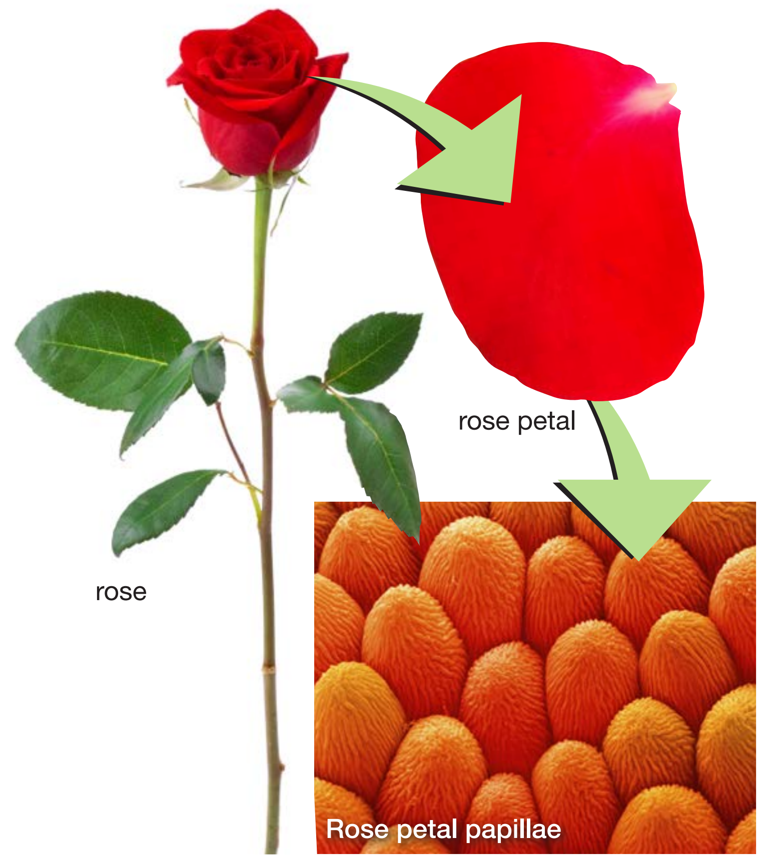
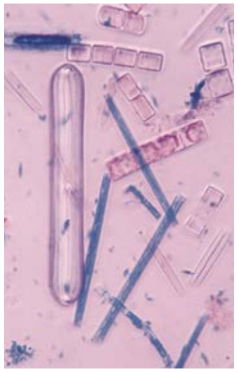
Introduction

Suppose you had microscopic X-ray vision, and you took a deep look inside living things. What would you learn? What would you discover deep inside a rose, for example? Or an elephant? Or a human being like yourself? This book will explore these questions by looking deep inside living things.

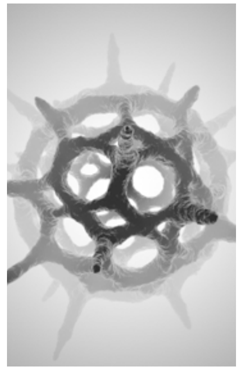


The closer you look at an organism, the more structures and parts you see.

algae



diatoms



protozoa

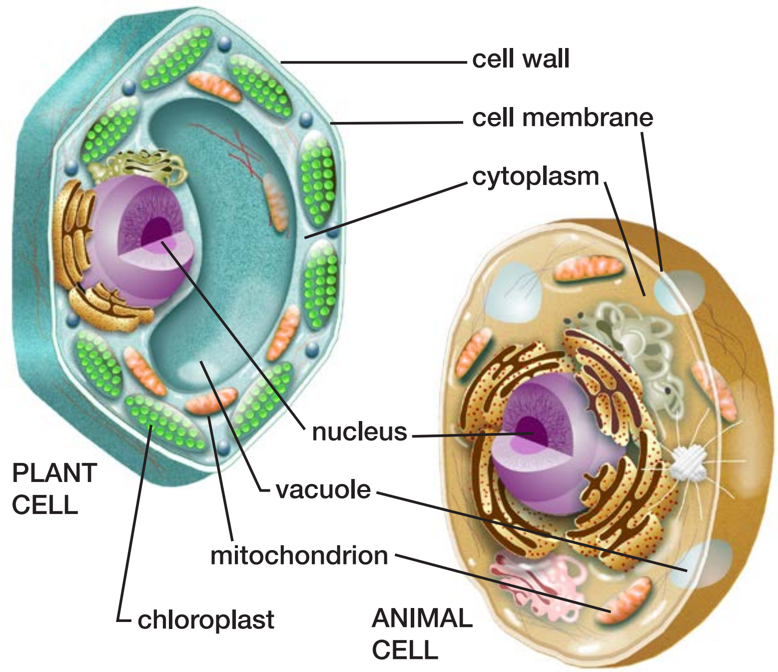
Cells

Living things share one basic characteristic: They are all made of **cells**. Cells are the smallest units of life. They produce the energy that makes living things grow and function.

Some living things have only a single cell. Called **unicellular** organisms, they include tiny algae, diatoms, and protozoa. Each of these single-celled organisms carries on all the activities it needs for survival within its single cell. These organisms live in a wide range of places, including ponds, rivers, soil, and even inside the bodies of other living things.

Other organisms large and small, from fleas and mosses to whales and giant redwoods, are **multicellular** organisms. They are made of trillions of cells that work together. The bigger the organism, the more cells it usually has.

There are two basic types of cells: plant cells and animal cells. Let's look at a typical animal cell. The outer covering of an animal cell is called the **cell membrane**. The cell membrane is a sac with **pores**, or small openings. This membrane is like a filter. It allows some things to pass in and out of the cell. The cell membrane allows nutrients to enter the cell and waste to exit. Inside the cell membrane is a jellylike fluid called **cytoplasm**. Other cell parts float in the cytoplasm. One important part is the **nucleus**. The nucleus directs the cell's activities, much as your brain controls your activities.



Plant and animal cells have many of the same parts. But plant cells have a cell wall and chloroplasts, which animal cells don't.

PLANT CELLS: FOOD AND STORAGE

Animals have to eat to survive, but plants can make their own food. Plants store a special substance called *chlorophyll* in tiny structures called *chloroplasts*. Animal cells do not have chlorophyll or chloroplasts. A plant uses chlorophyll to make food from water and a gas in the air. The plant absorbs sunlight and uses the Sun's energy to make food from the water and air. This process is called **photosynthesis**.

In many ways, plant cells are like animal cells. Both have a cell membrane, a nucleus, and cytoplasm. But they also have some parts that are different. A typical plant cell has a **cell wall**, which surrounds the cell membrane and gives the cell support. Animal cells do not have cell walls. The support that a cell wall gives the plant cell is important because plants do not have a skeleton for support.

All cells turn food into energy. Cells have hundreds or even thousands of **mitochondria**, which change energy into forms that the cells can use to do their jobs.

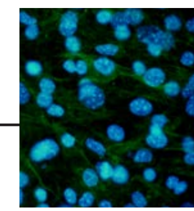
Most cells also have several sac-like structures called **vacuoles**, which store food and water until the cell needs them. They also hold waste until it can be removed from the cell.

Different Kinds of Cells

Not all cells are alike. An organism's cells differ, depending on their function. To better understand how cells can differ, let's take a look at some kinds of cells found in mammals.

Highly complex living things like mammals and reptiles have many parts that work together. Each part is made up of different kinds of cells. These animals have muscle cells, brain cells, nerve cells, blood cells, skin cells, and hundreds of other types of cells. All these cells must work together to keep the body growing and functioning.

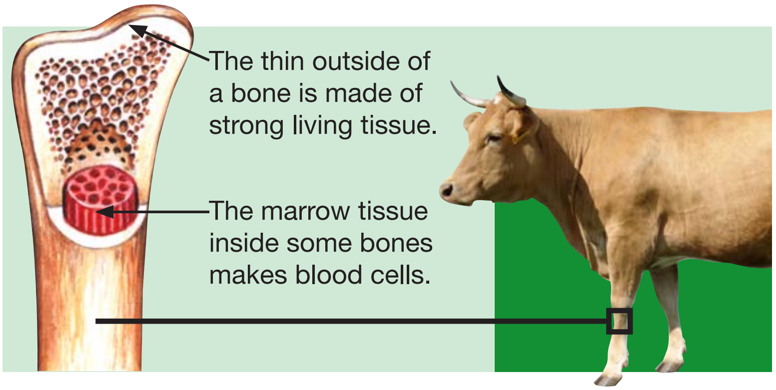
Do You Know?



How many cells are in your body? There is no way to know the exact number. How could we count every single one? The number varies, depending on the size of the person, and the number is constantly changing. Every second you are losing cells, but your body is also manufacturing new ones.

The best that scientists can do is estimate, and most say that the average human body contains about 75 trillion (75,000,000,000,000) cells.

To get a better idea of how tiny a cell is, think about this. It would take ten thousand (10,000) cells just to cover the head of a pin.



Tissues

Similar kinds of cells join together as **tissues** to perform the same task. For example, the muscles that animals use to move arms and legs are made of muscle cells. Other kinds of tissue make up bones, skin, blood, and nerves. Since you are most familiar with your body, let's take a closer look at some human tissues.

Bone is made up of two kinds of tissue. The outside of bones is made of tissue that contains the mineral calcium. Calcium makes bones hard

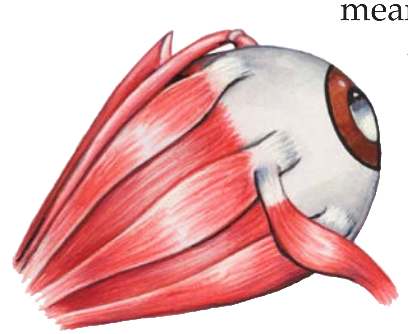
PLANT TISSUES

Both plants and animals have tissue, but plant tissues are different from animal tissues. Plants have *xylem* (ZY-lum) tissue for transporting water, and *phloem* (FLOW-um) tissue for transporting food.

to provide protection to softer parts of the body and support for muscles. Inside larger bones is another tissue called *marrow*. This tissue manufactures blood cells.

Your tissues are always working to keep you alive. Muscle tissue in your heart is constantly pumping blood tissue through your body. Blood tissue carries nourishment to cells and carries waste from cells. The muscle cells in muscle tissue contract, shorten, and then relax to move different parts of your body. You could not read these words without muscle tissue. Right now, muscle tissue is moving your eyes back and forth across this page.

At the same time, nerve tissue carries messages from cell to cell, directing your eye muscles to move and carrying signals back to your brain. Your brain is a mass of nerve tissue that interprets the words you see and gives them meaning. Your muscle tissue and nerve tissue are cooperating right now. They are working together to help you read these words.

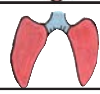
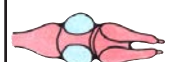
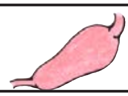
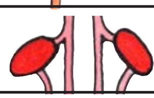


The muscle cells in this dog's eye contract and then relax to move the eye back and forth.

Organs

When different types of tissue work together to perform a special function, they form an **organ**. Eyes are organs. So are lungs, brains, bones, skin, and hearts. Animals as small as grasshoppers and as large as whales have many of the same organs, such as a heart, brain, and stomach.

The table below shows various frog organs and the special function they perform.

Frog Organs	Art	Function
Heart		pumps blood to tissues
Lungs		put oxygen into the blood
Brain		directs all body activities
Stomach		breaks down food
Small intestine		absorbs nutrients from food
Kidneys		store waste from body
Liver		cleans waste from blood

PLANT ORGANS

Both plants and animals have organs. Plant organs include the flower, stem, leaves, and roots. What do you know about each organ's purpose?

Body Systems

This book has explained that cells combine to form tissues, and tissues combine to form organs. When several organs work together in an animal, they form a bigger and more complex unit known as a **body system**. This is where all the work done by cells, tissues, and organs comes together to keep an organism alive and growing. Every animal has body systems.

For example, in an animal like a frog, several major systems work together to keep the frog alive. Each system has a special job to perform. For instance, the job of the digestive system is to break down the food that the frog eats so its cells can make energy from it. All the systems contribute to the frog's survival. The chart on the next page identifies ten systems found in frogs and other animals, including humans. Notice how each system does something unique and vital.

PLANT SYSTEMS

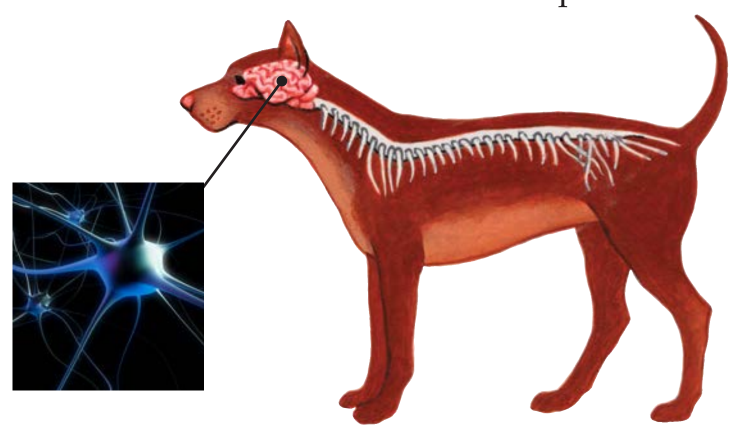
Plants have systems, too. They have the *root system*, below the ground, and the *shoot system*, which is everything above the ground.

TEN MAJOR BODY SYSTEMS		
System	**Main Organs**	**Activities**
Nervous	brain, spinal cord, nerves, and sense organs	Directs body's physical and mental activities
Muscular	frog: more than 60 muscles human: more than 600 muscles	Moves the body and substances within the body
Skeletal	frog: 159 bones human: 206 bones	Protects and supports the body
Respiratory	lungs and trachea	Supplies oxygen and removes gas wastes
Digestive	mouth, esophagus, stomach, intestines, kidneys, and liver	Breaks down food so the body can use it
Circulatory	blood vessels and heart	Transports oxygen, nutrients, and cell wastes through the body
Endocrine	pancreas, thyroid, pituitary, and other glands	Controls activities of internal organs
Immune	white blood cells and antibodies	Defends the body against bacteria and other invaders
Reproductive	female: ovaries, uterus male: testicles	Makes it possible for organisms to produce new offspring
Excretory	liver, kidney, bladder, lungs, skin	Removes waste from the body

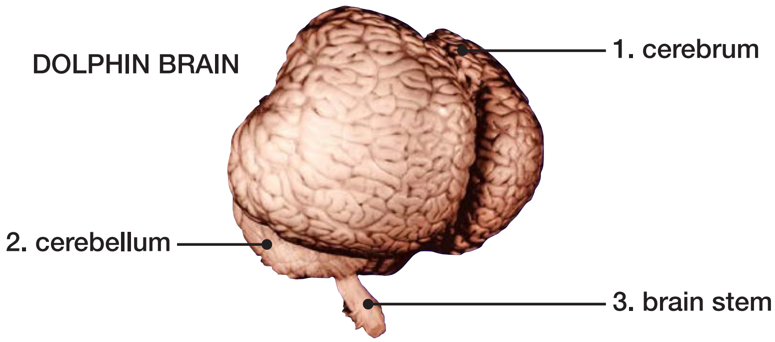
Mission Control

All our systems are vital, but the nervous system could be considered the communication and control system for all the other systems. It gathers information and processes this information to control almost all of an organism's functions. Nerve tissues in the sense organs—nose, eyes, ears, skin, and tongue—are part of this system. So are the bundles of nerves that connect the brain, sense organs, and other body parts. (Plants do not have nervous systems. Instead, their cells communicate by using chemicals that flow through the tissues.)

These nerve cells differ from other types of cells in many ways. Most importantly, they conduct electrical signals that allow body parts to communicate with each other. Sense organs use nerve cells to send information to the brain, and the brain sends back orders to respond.



Nerve cells in this dog's brain interpret information from the rest of its body.



The Boss

The brain of some animals has three main parts, and each plays a unique role in directing the body's activities. (1) The cerebrum processes incoming information from the senses and helps control thinking and speaking. It is the largest part of the human brain. Most animals' brains have smaller cerebrums than human brains. (2) The cerebellum is in charge of controlling posture and balance. It helps coordinate muscle movement. (3) The brain stem controls the circulatory, respiratory, digestive, and other systems.

Do You Know?

Other animals have brains, too. Dogs, rats, lizards, and grasshoppers have brains. Even tiny ants and fleas have brains. A giant blue whale's brain is four times bigger than a human's brain. But even though the whale brain is large, it is not as big compared to the size of its body as a human brain is.

Tears and saliva help defend the body.

Do You Know?

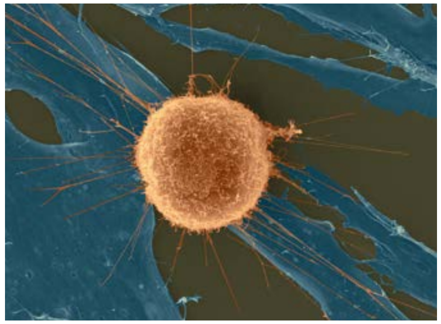
Not all microorganisms are harmful. In fact, many are beneficial. For example, tiny bacteria in your digestive system help you digest the food you eat. Foods like yogurt contain these helpful bacteria.

Body Wars

Did you know that most organisms have a special system devoted to protecting them from other tiny, but harmful, organisms? This is called the *immune system*. Every day, animals and plants are threatened by harmful bacteria, viruses, molds, fungi, protozoa, and other microorganisms that can cause disease. These microorganisms are known as **pathogens**.

Pathogens can enter an organism every day through its various openings. The organism works to keep out pathogens. The skin of animals and plants stops many of them. The saliva in humans' and other animals' mouths, as well as the digestive juices in the stomach kill some. Even tears wash some pathogens away.

Some pathogens still succeed in breaking through outer defenses and threatening an organism's health. They can infect cells and spread disease. At that point, an organism's immune system enters the battle. In animals with blood, white blood cells recognize the invader and quickly make more white blood cells. This large group of white blood cells attacks the pathogens and kills them, along with the cells they have infected.


A human white blood cell attacks cancer cells.

Meanwhile, other white blood cells produce **antibodies**—chemicals designed to kill certain kinds of pathogens. The antibodies cause the pathogens to clump together. As clumps, they are easier to recognize and destroy.

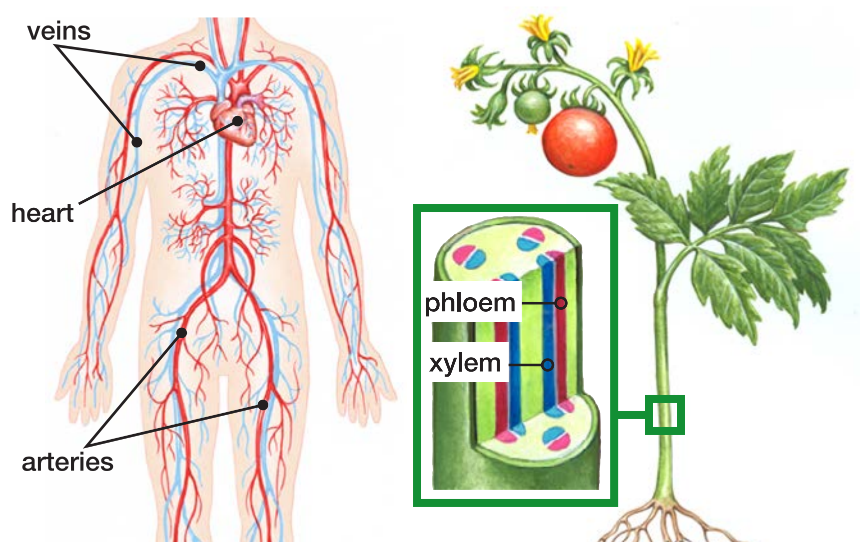
Do You Know?

All organisms have some way of defending themselves against disease. Trees ooze sap to heal cuts and keep away invasive bugs. Plants and insects use chemicals to ward off microorganisms. Even tiny bacteria use special parts to engulf and destroy other bacteria.

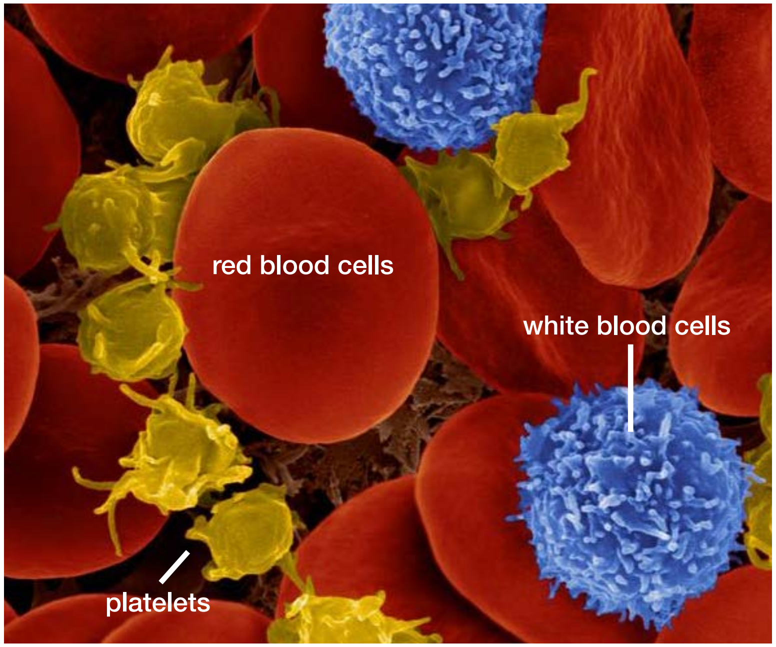
Rivers of Nutrients

Let's take a closer look at one other important system—the circulatory system. This system is like a vast series of interconnected tubes. In both animals and plants, the liquid running through these tubes delivers nutrients and other substances to cells. It also hauls away the cells' waste products. In animals, the liquid is blood, and in plants, it is mostly water.

In animals, the tubes are miles and miles of blood vessels. Vessels called **arteries** carry blood to the cells, while vessels called **veins** carry blood away from cells. Arteries and veins are connected by tiny vessels called **capillaries**, which deliver antibodies to cells and take waste products from cells.



In humans, blood vessels circulate nutrients to all parts of the body. In plants, the phloem does this job.



In mammals, red blood cells have no nucleus, but white blood cells do. Platelets are not cells — they are packets of proteins.

The liquid part of animal blood is called **plasma**. Plasma is mostly water, but it also carries red blood cells, which carry oxygen to body cells, and white blood cells, which help fight off diseases. Blood also contains platelets, which help clot blood and stop bleeding when you are injured.

Do You Know?

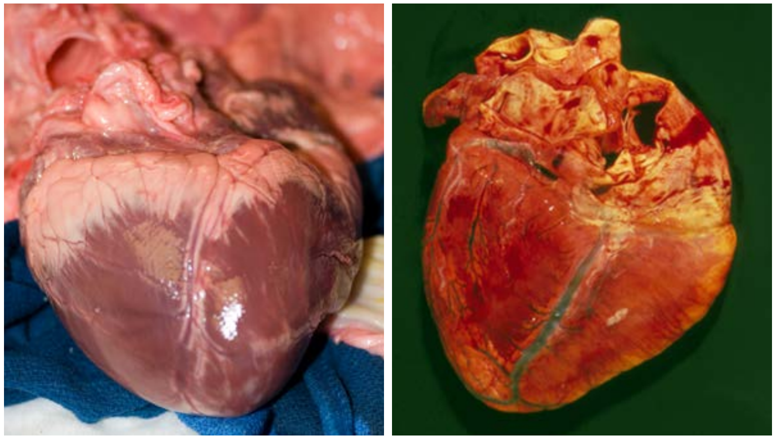
Insects have blood, too—but it's green instead of red. That's because insect blood doesn't carry oxygen. Insects have tiny holes in their bodies that let in oxygen and release waste gas.

Blood has one more vital task to perform. Not only does it bring oxygen to the cells from the lungs, but it also hauls away a gas called carbon dioxide (CO_2) and other waste products. Cells create this waste when they convert food and oxygen to energy.

Breathing out releases waste gases from your cells.

The blood carries the CO_2 back to the lungs, where you release it when you exhale. Other wastes are delivered to the kidneys and are released when you go to the bathroom. In plants, the xylem tissue is made of cells and vessels that move water from the roots up to the leaves. Phloem moves nutrients made by the leaves to other cells. The sap, which usually comes from phloem, helps trees fight infections and heal wounds. Plants release waste through their leaves.

If the circulatory system were a factory, it would get high marks for cleanliness, neatness, and efficiency.



A pig heart (left) is similar in size and shape to a human heart (right).

In animals, the heart is the engine that runs this highly efficient factory. The heart is a powerful muscular organ about the size of your fist. An adult heart beats about 70 times a minute (it's a little faster in teens and children). Each beat moves blood along inside all those vessels, delivering nutrients to and removing waste from cells throughout the body as it performs its life-giving tasks.

Math Moment

If a human lives to the age of 80, and his or her heart beats an average of 70 times a minute, how many times will his or her heart beat through a lifetime? Hint: How many minutes are there in an hour, day, and year with 365 days?

Conclusion

Multicellular organisms are very complex machines. The human body alone has trillions of cells making up many different types of tissues that combine to make dozens of organs organized into systems that all work together.

Other living things have more or fewer cells, tissues, and organs, but they all must work together so each organism can live. No wonder scientists and doctors often specialize in one or another type of cell. Understanding the complex system that is your body will help you enjoy a long, healthy life.

Scientists study the structures inside living things to better understand how whole organisms can be their healthiest.

Glossary

antibodies	chemicals produced by the body that attack invading germs (p. 17)
arteries	blood vessels that move oxygen-rich blood from the heart to the body tissues (p. 18)
body system	a group of organs in the body that work together (p. 12)
capillaries	the tiniest blood vessels that carry blood from arteries to veins (p. 18)
cells	the smallest independently functioning units in an organism (p. 5)
cell membrane	the outside covering of an animal cell that is also inside the cell wall of plant cells (p. 6)
cell wall	the firm outside covering of a plant cell (p. 7)
cytoplasm	the liquid that lies inside the cell membrane (p. 6)
mitochondria	cell parts that produce energy (p. 7)
multicellular	made up of more than one cell (p. 5)
nucleus	the cell part that controls a cell's activities (p. 6)

organ	a part of an organism, made of different kinds of tissue, that has a specific function (p. 11)
pathogens	unicellular organisms that cause disease (p. 16)
photosynthesis	the process that plants use to make their own food (p. 7)
plasma	the liquid part of blood in which blood cells flow (p. 19)
pores	tiny openings (p. 6)
tissues	groups of cells in an organism that are similar in form and function (p. 9)
unicellular	composed of one cell (p. 5)
vacuoles	cell parts that store water, nutrients, and waste (p. 7)
veins	blood vessels that carry blood back from the body's cells to the heart (p. 18)

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